

# Noah Qiao

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## TECHNICAL SKILLS

**Mechanical:** SolidWorks, AutoCAD, 3D Printing, Sheet Metal Design, Engineering Drawings, Machine Tools

**Electrical / Hardware:** Altium, KiCad, LTspice, Falstad, PCB Design, Soldering, Sensor Integration, ESD & Surge Protection

**Embedded / Controls:** Arduino, ESP32, STM32, Raspberry Pi, Teensy 4.1, RTOS, PLC Programming, VHDL

**Programming:** Python, C++, MATLAB, Java, SQL

## RELEVANT EXPERIENCE

### Borrum Energy Solutions

Sep 2025 - Dec 2025

R&D Electrical - Mechatronics Engineering

- Designed wind-turbine temperature/humidity monitoring unit with SD logging and **RTOS** task scheduling using **Teensy 4.1**.
- Designed **KiCad PCB** with surge and ESD transient protection for harsh outdoor deployment.
- Collaborated with UofT and UBC on microwave-based ice detection testing for next-generation wind turbines.
- Automated electronic component inventory using **Python** and vendor APIs, reducing manual entry by ~**85%**.

### Excelitas

Jan 2025 - Apr 2025

Mechanical Engineering Co-op

- Led assembly and validation of a new-generation power testing station, supporting design approval and production readiness.
- Improved fixture design, increasing testing efficiency by **300%** and reducing production time.
- Created sheet metal part models and manufacturing drawings for product lines using **SolidWorks**.
- Revised PCB connector layouts and schematics using **Altium** to support updated product requirements.

### Sheartak Tools Ltd

Sep 2023 - Dec 2023

Mechanical Engineering Associate

- Designed 10+ **SolidWorks** cutter-head profiles for customer machine upgrades.
- Produced detailed engineering drawings to expand inventory and streamline manufacturing.
- Supported machine upgrade work that improved woodcutting quality and customer usability.

### Magna International

Jan 2023 - Apr 2023

Test-Technician

- Conducted tests on 10,000+ customer-supplied products per specified test methods.
- Organized and recorded test measurements in Excel TADs for future reference.
- Ensured testing measurements met ISO 17025 requirements and reported all outliers.

## SELECTED ENGINEERING PROJECTS

### Autonomous Search & Rescue Robot

Apr 2026

- Built autonomous S&R robot for line following, LEGO detection, pickup, transport, and return-to-start.
- Integrated Raspberry Pi vision, Arduino control, IR sensing, servo actuation, and serial communication.
- Designed vacuum pickup and modular 3D-printed chassis, achieving **80%** course completion and **90%** pickup success.

### Remote Control Battle Bot

Mar 2025

- Designed and built a remote-controlled combat robot with an integrated spinning weapon system.
- Integrated FlySky FS2A receiver and dual-channel brushed ESC to enable wireless control.
- Conducted soldering, wiring, and troubleshooting of low-voltage DC circuits, signal connectors, and motor systems.

### Gesture Controlled Robotic Hand

Aug 2024

- Designed and 3D printed **SolidWorks** components for a gesture-controlled robotic hand.
- Developed **Arduino Uno** motor control for real-time finger actuation.
- Implemented gesture tracking using **Python**, **MediaPipe**, and **OpenCV**.

## EDUCATION

### University of Waterloo

Sep 2022 - Apr 2027

Candidate for Bachelor of Applied Science in Mechatronics Engineering